

Problem 28.1

$$J M = \begin{bmatrix} m_{21} & m_{22} & m_{23} & m_{24} \\ 0 & 0 & 0 & 0 \\ m_{41} & \cdot & \cdot & m_{44} \\ 0 & 0 & 0 & 0 \end{bmatrix} \quad M K = \begin{bmatrix} 0 & m_{11} & m_{12} & 0 \\ 0 & m_{21} & m_{22} & 0 \\ 0 & \cdot & \cdot & 0 \\ 0 & m_{41} & m_{42} & 0 \end{bmatrix}$$

Problem 28.2

a) Polynomials of A & B are similar.

b) $A = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}$ $B = \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}$

c) $M = \begin{bmatrix} 1 & -1 \\ 0 & 1 \end{bmatrix}$

d) different number of eigenvectors.

e) $M = \begin{bmatrix} 0 & 1 & 0 & 0 \\ 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$